

Industry Problem Task – Project Report

INDUSTRY PROBLEM TASK – AI/ML PROJECT REPORT

1. **ABSTRACT** This project develops an educational end-to-end AI/ML workflow for diabetes risk prediction and a simplified reinforcement-learning treatment recommendation simulator. It covers inductive learning, class imbalance, decision trees, pruning, logistic regression, feature importance, MDP formulation and Q-learning.

2. **INTRODUCTION** Healthcare AI can support early risk stratification when models are trained and evaluated carefully. The prediction problem is treated as supervised binary classification. The treatment component is formulated as an abstract MDP to demonstrate reinforcement learning without claiming clinical treatment validity.

3. **DATA PREPARATION** Features: Age, BloodPressure, Cholesterol, Glucose, BMI and FamilyHistory. Target: Diabetes. A stratified 70:30 split is used. SMOTE is applied only to the training set. Standardization is applied to Logistic Regression; tree models do not require normalization.

4. **DECISION TREE** A Gini-based Decision Tree is trained. Pre-pruning uses max_depth and min_samples_leaf. A second model uses cost-complexity post-pruning. Accuracy, Precision, Recall, F1 and the confusion matrix are reported by Python_Code.py. False Negatives represent missed diabetic cases and are especially important for screening. Recall/sensitivity, threshold tuning and class-aware training can reduce them.

5. **STATISTICAL LEARNING** Logistic Regression is used as a statistical baseline. AUC-ROC is included because it evaluates ranking quality across thresholds. Absolute standardized coefficients provide a comparable feature-importance view. Agreement between the tree and logistic model should be interpreted carefully because the models capture different relationships.

6. **REINFORCEMENT LEARNING MDP** States: Stable, AtRisk, HighRisk, Improving. Actions: Diet, Exercise, Medication, Monitor. Rewards represent an abstract health-improvement score. Transitions define how actions change the state. Q-learning uses $\alpha=0.5$, $\gamma=0.9$ and $\epsilon=0.15$ for at least five episodes.

7. **Q-LEARNING UPDATE** $Q(s,a) = Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$ The final policy selects the action with the highest Q-value for each state. The program prints updates across two iterations and produces a final Q-table.

8. **ETHICAL CONSIDERATIONS** A medical RL system must not freely explore treatment actions on real patients. Safety constraints, clinician approval, bias testing, explainability, privacy, uncertainty estimation, monitoring and external validation are required. The reward function must not encourage unsafe shortcuts. The model must be treated as decision support rather than an autonomous prescriber.

9. **LIMITATIONS** No real patient dataset was supplied, so the Python program creates a reproducible synthetic dataset. Therefore numerical metrics are demonstrations of the workflow, not clinical evidence. The MDP is also synthetic and should not be interpreted as medical advice.

10. **CONCLUSION** The project integrates supervised learning and reinforcement learning into one healthcare AI/ML case study. The pipeline addresses data imbalance, interpretable diagnosis, statistical risk modelling, feature importance and Q-learning while emphasizing the safety requirements of medical AI.

11. **FUTURE ENHANCEMENTS** Use a validated real-world dataset; cross-validation; calibration; external testing; subgroup fairness analysis; explainable AI; temporal patient records; offline/constrained RL; clinician-in-the-loop approval; uncertainty estimates; and prospective clinical validation.